Studio North: is growth becoming more valuable?

Customer acquisition, monthly renewal and contribution economics | 24 monthly acquisition cohorts, 2024-01 through 2025-12; data cutoff 2026-01-01

THE BUSINESS AND THE DECISION

Studio North sells a monthly creative-work membership to independent professionals. Standard and premium plans combine ready-to-use templates with guided learning; members can also purchase workshops and other add-ons. Each month, members choose whether to renew. Search campaigns and partner workshops bring customers with two stated priorities: save time or build skills.

Management needs to know whether successive acquisition cohorts create more durable contribution, at a cost the company can recover quickly enough to fund growth. Company scale, customer quality and marketing incrementality answer different parts of that decision.

The company is moving toward stronger customer economics: lower blended CAC, higher contribution and more cash recovered over six months. Later cohorts also retain more customers at common ages, including within each channel–need group. Short campaign tests do not yet establish the return from a larger acquisition budget.

ACQUISITION EFFICIENCY
$109 → $81
blended CAC per acquired customer

CUSTOMER VALUE
$110 → $132
six-month contribution per customer

CASH RECOVERY
$2 → $51
six-month cash after blended CAC

Matched-age comparison: January–June 2024 versus January–June 2025; age 5, six observed receipts; acquired-customer weights. Values above are rounded to dollars; the evidence table reports cents.

WHAT THE EVIDENCE ESTABLISHES

Table 1. Matched-age evidence on acquisition efficiency, customer value and cash recovery.

Evidence	Earlier	Later	Interpretation
Sixth-month retained share	54.3%	63.3%	Observed at age index 5
Blended CAC	$108.79	$81.44	Tracked spending / all acquired people
Six-month contribution / customer	$110.33	$132.50	After recorded variable costs
Six-month cash after acquisition	$1.53	$51.06	Contribution less blended CAC
Ex-test CAC / cash	$108.79 / $1.53	$85.66 / $47.51	Same joining windows; both test arms excluded

Reported economics include experimental buyers. The later window contains 86 test-assigned buyers and $650 of test spending; excluding both arms and their contribution gives the sensitivity row. The earlier window excludes 0 buyers. Improvement remains; its magnitude is smaller.

RECOMMENDED MANAGEMENT RESPONSE

Action. Marketing and Analytics should preserve controls and extend both campaign follow-ups through the cash-recovery horizon. Finance should monitor six-month cash against the ex-test matched window's $47.51 per customer, not use it as an automatic causal gate. Product should protect the experience supporting stronger renewal and monitor whether the newest cohorts sustain it as acquisition scales. Incremental contribution after media per assigned prospect: search / save time: -$0.65 over 31 days (95% interval -$2.29 to $0.98); partner / build skill: -$1.31 over 30 days (95% interval -$3.13 to $0.51). 2 of 2 intervals include zero; neither the sign nor the ranking of point estimates warrants a scale-or-stop decision.

WHAT WOULD CHANGE THE DECISION

For immature cohorts, value depends on future renewals and spending. Treat their LTV as a conditional forecast. Scale a campaign only when a randomized comparison supports incremental contribution over a relevant window; historical attributed CAC alone cannot establish that return. Retain the cost and cash-recovery constraints when a campaign produces more revenue.

All customer records describe an illustrative business and are synthetic. The analysis covers recorded variable costs and acquisition spending, not fixed operating costs, financing or taxes. It therefore does not establish company profitability.

The management pages summarize the diagnosis and action. Figures 5–6 connect the customer journeys to evidence. Tables 11–18 classify all eight raw sources using the Universal Customer View. The appendices document parameters, validation and reproducibility.

Compare customer quality at a common age

A joining cohort contains customers whose first successful payment occurred in the same month. Future ages remain unobserved.

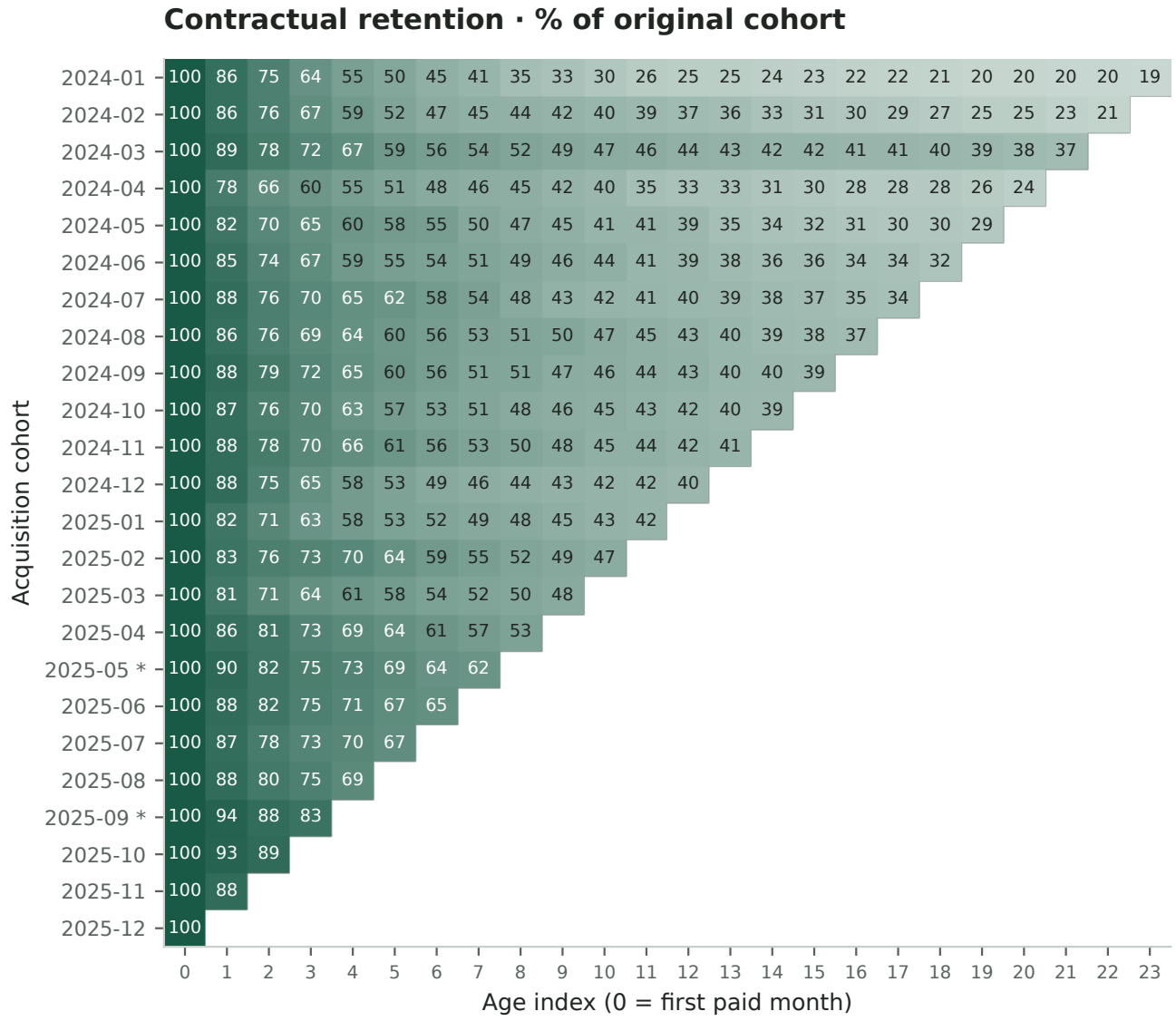


Figure 1. Customer retention for all 24 monthly acquisition cohorts and every observed age. Cells are percentages of the original cohort; blanks are unobserved. * marks test months, including buyers from both arms.

Saved SQL output: `cohort_totals.csv`. The full age range is 0–23; age 0 is the first paid month. Read across a row to follow one cohort and down a column to compare successive cohorts at the same age. No missing future month is assigned zero retention.

SEPARATE COHORT AGE FROM CALENDAR TIME

Every cohort starts at 100% because acquisition defines its original denominator. Newer cohorts have shorter histories. Their shorter lines and blank cells do not establish shorter customer lifetimes. Figure 3 shows the same full cohort sequence as curves; Figure 7 recombines cohorts in calendar time.

An improving business should acquire cohorts that retain more customers at comparable ages, rather than relying only on a larger inflow of new customers.

Follow contribution through acquisition recovery

All 24 cohorts | Cash balance uses recorded contribution, tracked acquisition cost and the original acquired-customer denominator.

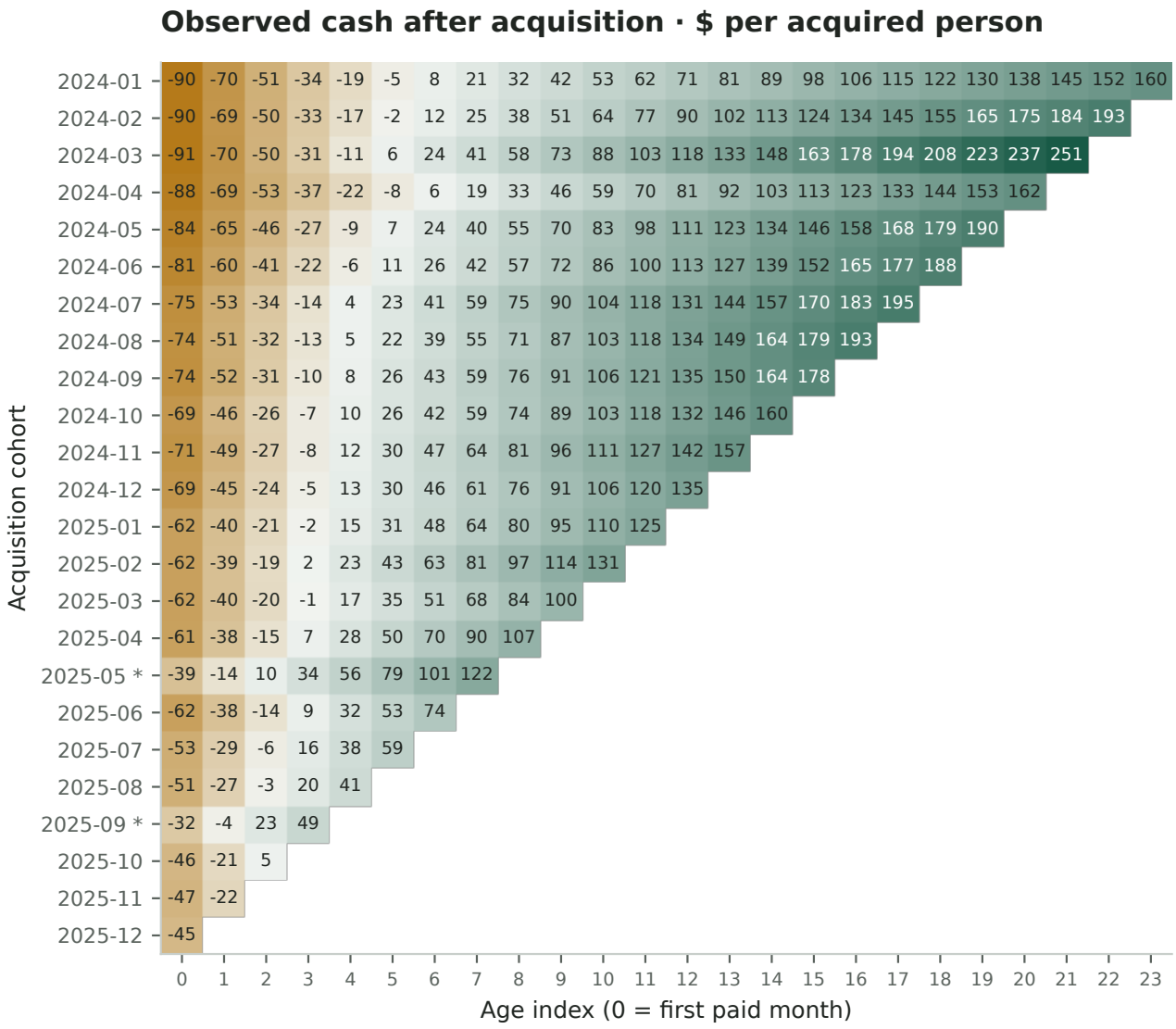


Figure 2. Cumulative cash after acquisition for every joining cohort, in dollars per original acquired customer. Orange indicates unrecovered cost; green indicates a positive balance. * marks test months, including both arms.

Saved SQL output: `cohort_cash_economics.csv`. Each cell equals cumulative net revenue less incurred variable costs and tracked acquisition spending, divided by original cohort size. The blank region is unobserved. This is undiscounted cash, not total company profit or discounted LTV.

ACQUISITION EFFICIENCY AND DURABILITY MUST MOVE TOGETHER

At age 5, six-month cash after acquisition changes from $1.53 to $51.06 per original customer. The comparison uses January–June 2024 and the same joining months of 2025, weighted by acquired cohort size. Table 1 reports the matching retention, contribution and blended-CAC evidence. These are descriptive comparisons, not campaign effects.

> A new cohort may recover acquisition cost sooner because it costs less to acquire, generates more contribution per retained member, retains more members, or combines these changes. Use the retention and cash triangles together rather than treating either as a complete explanation.

Read the whole sequence, not a selected curve

Every monthly acquisition cohort is shown. Shared axes and one chronological color scale keep the comparisons aligned.

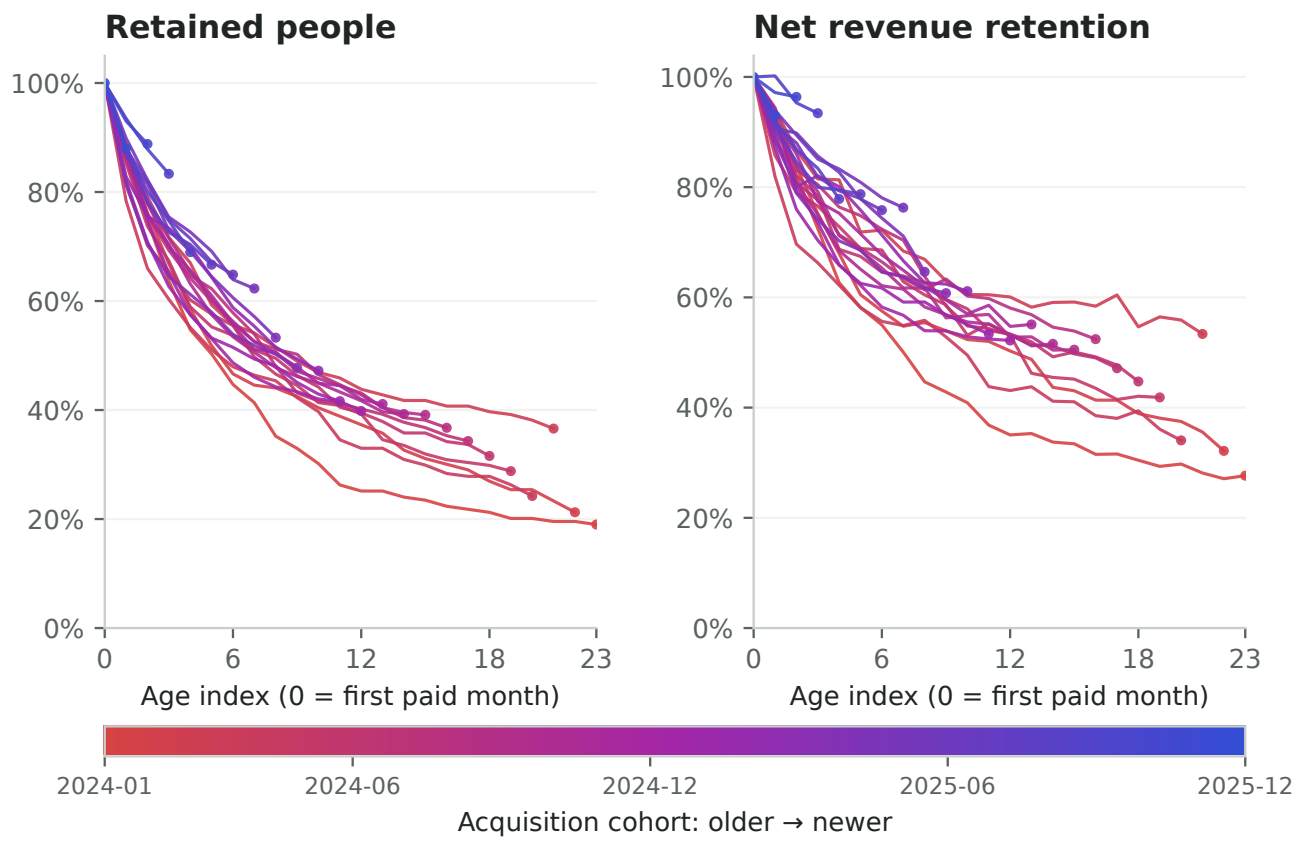


Figure 3. All 24 customer-retention and revenue-retention curves. Older cohorts are red and newer cohorts are blue; each line ends at its last observed month.

Retained share uses original cohort membership; revenue retention uses each cohort's first-period revenue. Revenue can change because of the number of renewing members, their plan mix or add-on spending. Product activity is not a substitute for a renewal record.

HOW TO READ MANY COHORTS

Compare the vertical position of differently colored curves at the *same* age. Later cohorts lying above earlier cohorts suggest improving retention; they need not dominate at every month. The chronological color key replaces an unwieldy 24-item legend. A curve with only its acquisition point has no observed renewal history.

CUSTOMER COUNTS AND REVENUE ANSWER DIFFERENT QUESTIONS

Higher revenue retention can coexist with customer losses when remaining members upgrade or buy add-ons. A retention improvement should therefore be checked against contribution and acquisition recovery in Figure 2, and against comparable customer groups in Figure 4.

The trajectory matters as well as the cutoff. A stronger first month does not establish longer customer life, while a customer-count decline can coexist with more spending by the members who remain.

Check whether progress survives the customer mix

Observed renewal trajectories by acquisition channel and stated need; each panel uses the same axes and all 24 cohorts.

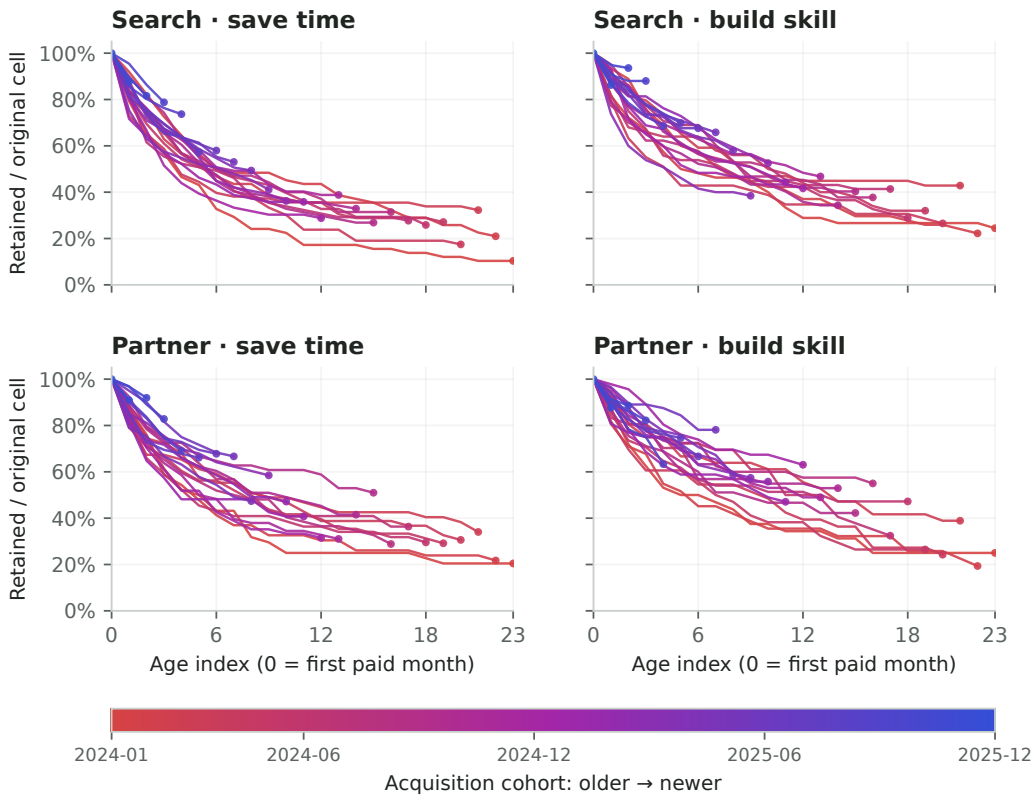


Figure 4. The full cohort sequence within each channel-by-need group. The color key identifies acquisition month. Paid channels are shown; organic test-control cells are omitted.

Saved SQL output: `cohort_age_summary.csv`. Curves divide retained members by the original cohort–channel–need cell. No missing future observations are filled with zeros.

WITHIN-GROUP DIAGNOSIS

Age-5 paid retention is 54.3% for January–June 2024 versus 63.4% for the same 2025 joining months. Fixing the earlier channel-by-need mix gives 54.3% versus 63.0%; 4 of 4 paid groups improve. This removes changes in these groups' shares, not all customer differences or confounding. All 24 trajectories are shown above.

CHECK HOW MUCH THE GROUPS ACTUALLY SEPARATE

The primary need grouping has the lowest within-group variance in six-month contribution (4,366.3 dollars squared), a 2.5% reduction versus the next-best scheme. This modest separation leaves residual heterogeneity: a need label does not make customers economically identical.

Table 2. Residual within-group variation in six-month contribution under candidate segmentation schemes.

Grouping	Variance (dollars squared)	Groups	Smallest group (people)
age band	4,480.4	2	2,130
primary need	4,366.3	2	2,007
region	4,480.5	2	2,087

Comparison uses the same mature-customer sample and the same six-month outcome. Smaller within-group variance means more homogeneous observed outcomes within the chosen groups; it does not imply identical customers or establish a causal effect of a need or channel.

Connect customer needs to the evidence we collect

Journey maps are hypotheses about experience. Events, relationship records and expressed perceptions provide different checks on those hypotheses.

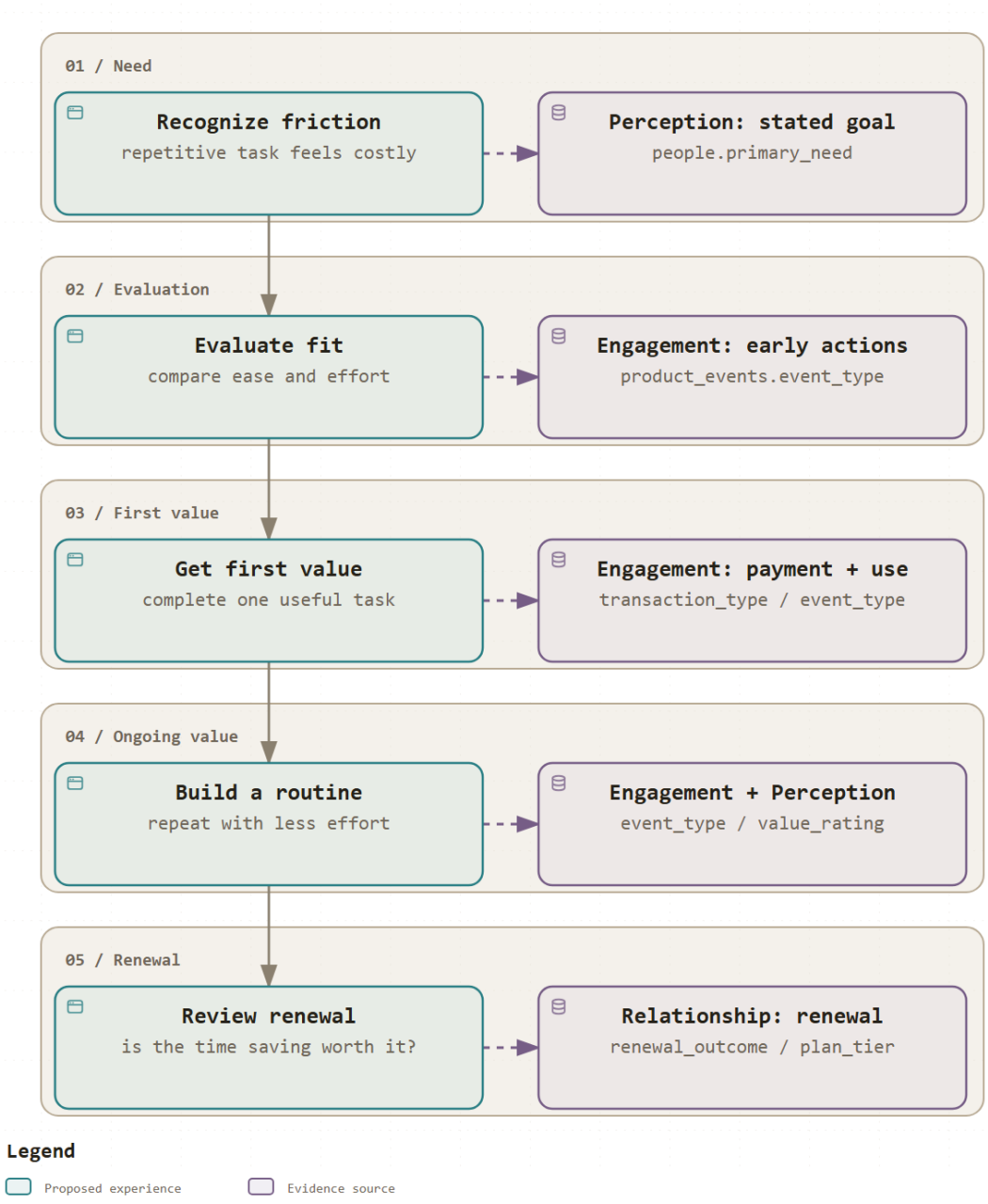


Figure 5. Save-time segment: a proposed path from work friction to repeat use and renewal, with the relevant evidence at each step.

The Universal Customer View separates identifiers, observed interactions, the commercial relationship and expressed perceptions. Tables 11–18 classify all eight raw sources and mark operational context explicitly.

A different need implies a different customer journey

The build-skill segment values learning progress. The same data sources can illuminate a different sequence of experiences.

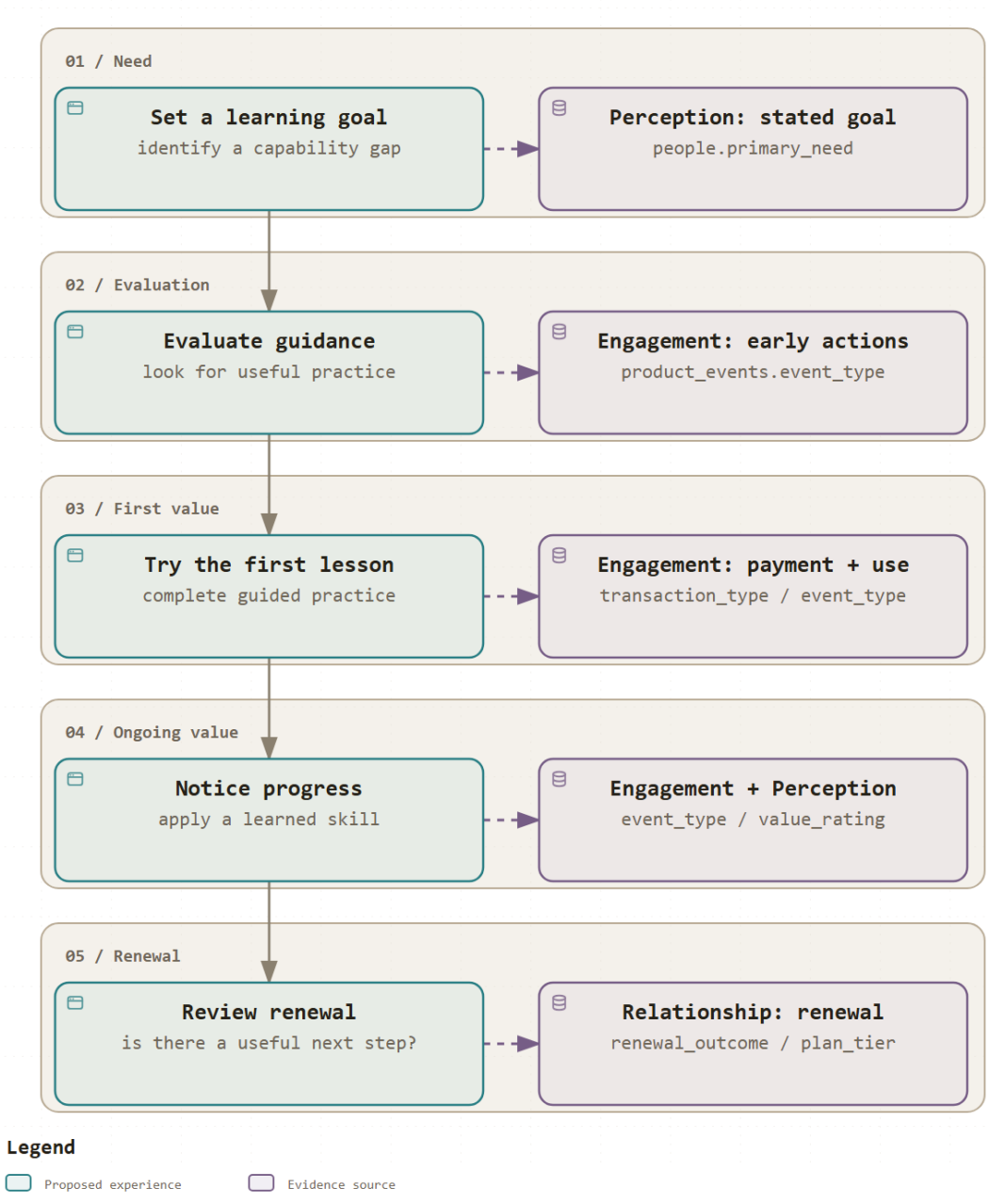


Figure 6. Build-skill segment: a proposed learning path and the evidence needed to assess progress, value and renewal.

The complete field-level classification appears in Tables 11–18. Missing perceptions remain missing; a click or renewal cannot establish what a customer believes.

Retained cohorts determine how growth accumulates

The company is the sum of its acquisition cohorts at different ages, not the latest cohort scaled up.

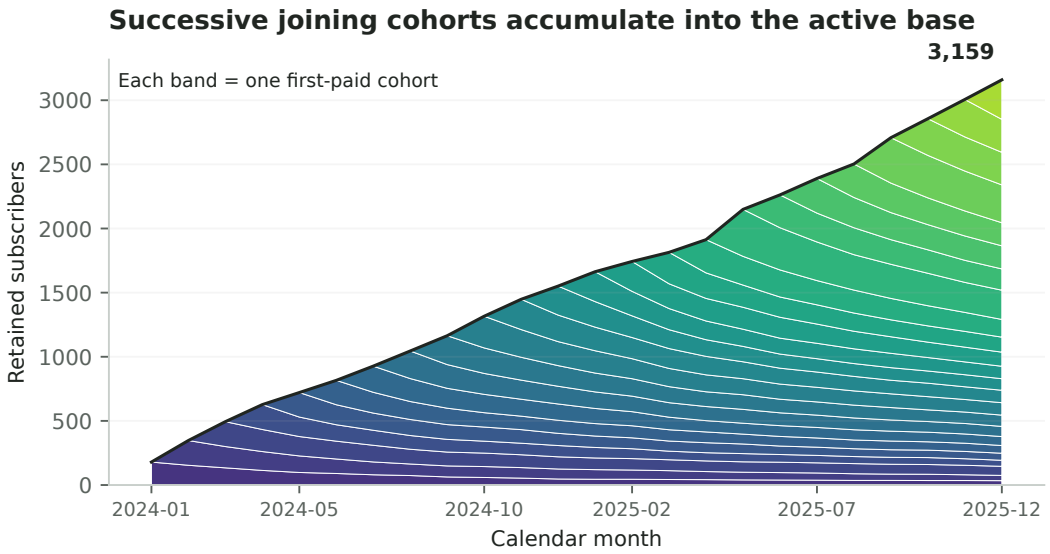


Figure 7. The company's retained membership base as the sum of all 24 acquisition cohorts in calendar time.

Saved SQL output: `company_month.csv`. Each layer is a joining cohort's active customers at the displayed calendar month. Layer totals are reconciled to independently saved company totals before plotting.

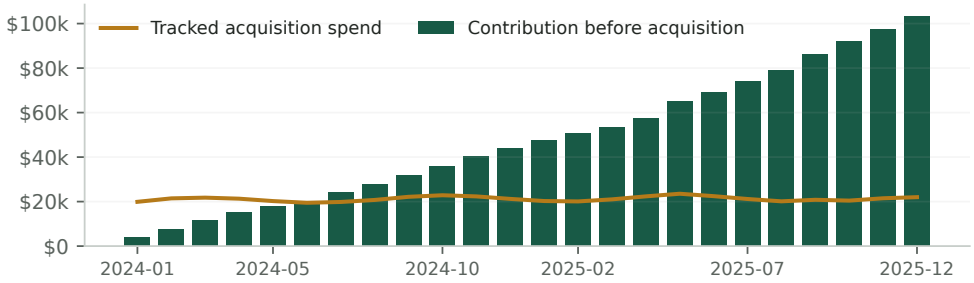


Figure 8. Monthly contribution before acquisition compared with tracked acquisition spending.

SCALE AND CASH RECOVERY ARE SEPARATE TESTS

New acquisition adds customers immediately. Renewals preserve older layers and their contribution, reducing the replacement acquisition required to sustain scale. The lower chart compares recorded contribution with acquisition spending in each calendar month; it is not cumulative retained customers or a profit statement. In 2025-12, recorded contribution was $103,375 and acquisition spending was $21,995, leaving $81,380 before fixed costs, financing and taxes. This period result should be read alongside the age-aligned cohort comparison, rather than used as a substitute for it.

OPERATING IMPLICATION

Track both the common-age economics of each joining cohort and the company's accumulated contribution. A favorable acquisition metric can coexist with slower cash recovery when renewals weaken. A growing customer base can also conceal deterioration if management replaces departing customers with increasingly costly new ones.

Cash payback is undiscounted: the first age at which cumulative contribution covers the acquisition investment. The LTV calculation that follows discounts expected monthly contribution and uses a common, explicit horizon.

Forecast monthly renewal before forecasting value

Hierarchical shifted-Beta-Geometric retention + Gamma gross revenue; NumPyro NUTS

HOW THE MODEL REPRESENTS THE BUSINESS

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$\text{net LTV}_H = \sum_{a=0}^{H-1} \frac{\text{E}[\text{contribution}_a \mid \text{observed history}]}{(1+d)^{a+1}} - \text{CAC}\,.$$

Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Keep observed revenue and contribution; forecast recursively from the last observed survivors. All cohorts are valued over the same 12 monthly receipts, discounting at 1% per month. Forecast bands cover parameter uncertainty, not renewal/revenue process noise or all future business risk. The main retention model extends Fader–Hardie's shifted-Beta-Geometric approach; the revenue model adapts Orduz's Gamma structure. An Orduz-inspired neural retention model is a separate validation benchmark, not the source of executive LTV. Parameter estimates and held-out checks appear in the appendix.

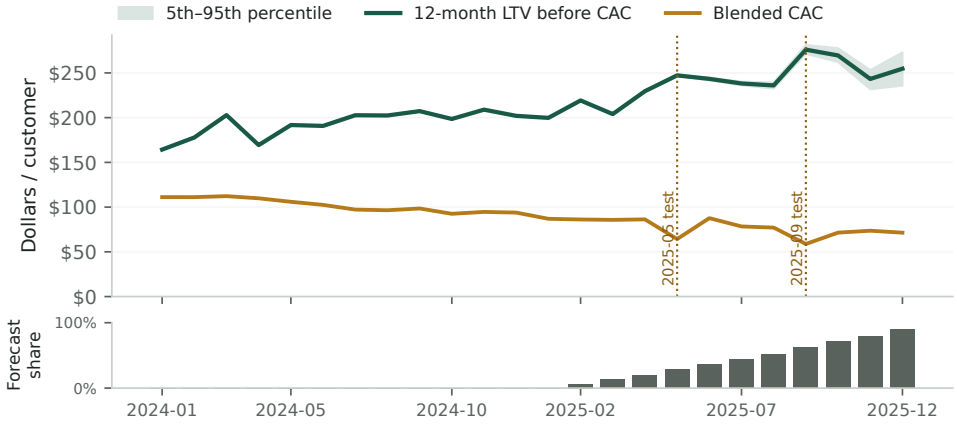


Figure 9. Common-horizon contribution LTV and blended CAC for all cohorts. The lower strip identifies the share of each LTV estimate that is forecast.

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval. The lower strip shows the forecast share of common 12-month LTV. Dotted markers identify experiment months with organic control buyers in the denominator. Blended CAC divides tracked spending by all acquired customers; it is not the causal cost of the next customer.

MANAGEMENT INTERPRETATION

Across the cohort sequence, common-horizon contribution LTV combines observed receipts with a growing predicted remainder; the strip makes that dependence visible. Net of blended CAC, the endpoint comparison is $53.15 for 2024-01 versus $183.30 for 2025-12. This is not a monotonic improvement claim. The newest value is 90% forecast, with a 5th–95th parameter range of $164.42 to $201.47.

Future net revenue and contribution shares of payment revenue use observed channel-by-need-by-age shares, with the same group's observed renewal ages (age>=1) as fallback. Shares are fixed assumptions, not causally identified or assigned posterior uncertainty. Predicted customer value is not experimentally identified campaign return. A favorable average LTV-to-CAC comparison still requires a cash-recovery constraint and evidence that the next acquisition investment creates additional value.

Require the next campaign to beat its own baseline

Randomized assignment includes all eligible prospects, including those who never purchase. Each test keeps its own control and follow-up window.

Table 3. Assigned prospects, buyers, recorded contribution and media spending by randomized arm.

Test	Arm	Assigned	Buyers	Contribution	Media
search / save time	control	250	31	$765.90	$0.00
search / save time	treatment	250	55	$1,252.50	$650.00
partner / build skill	control	250	31	$798.65	$0.00
partner / build skill	treatment	250	50	$1,370.25	$900.00

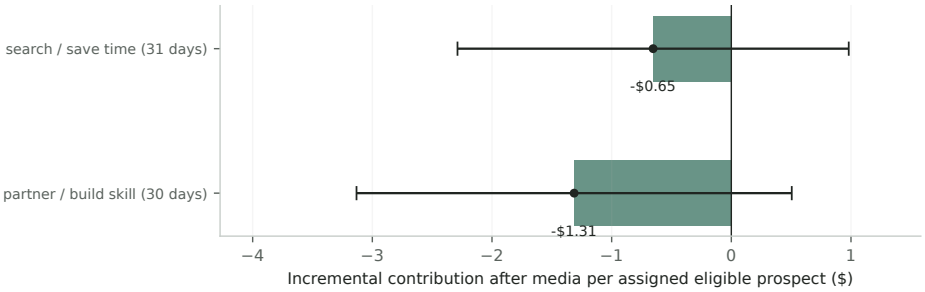


Figure 10. Incremental contribution after media per assigned prospect, with approximate 95% intervals over each test's recorded follow-up window.

Saved SQL outputs: `experiment_summary.csv` and `experiment_effects.csv`. Whiskers are approximate 95% intervals from all-assigned contribution variances, including nonbuyer zeros. Fixed media costs shift the mean, not its sampling variance. These 30/31-day windows are shorter than the cohorts' cash-recovery horizon; they do not measure lifetime campaign return.

THE SAME ECONOMICS APPLY TO TREATMENT AND CONTROL

For each arm, net revenue less recorded variable costs gives contribution. Compare contribution per assigned eligible prospect, then subtract the treatment's additional media cost. Control purchases remain in the baseline rather than receiving campaign credit.

$$\Delta\text{contribution after media per prospect} = \frac{C_T - M_T}{N_T} - \frac{C_C - M_C}{N_C}.$$

MANAGEMENT ACTIONS AND ACCOUNTABILITY

Table 4. Accountable teams, next actions and the evidence required to revisit the recommendation.

Accountable team	Next action	Decision evidence
Marketing + Finance	Defer expansion or rejection based only on the short-window profit estimates; monitor recovery against mature-cohort experience.	Age-aligned value and cash, alongside uncertainty in incremental returns.
Product + Operations	Monitor the experience behind each segment's renewal trajectory; investigate departures as acquisition scales.	Renewal events, product actions and expressed customer evaluations.
Marketing + Analytics	Preserve both controls and extend follow-up through the acquisition cash-recovery horizon.	Contribution after media per assigned prospect, including nonbuyers, with uncertainty.
Data + Analytics	Refresh the evidence from the source exports and review forecast misses before releasing the decision report.	SQL reconciliation, model holdouts, parameter output and file hashes.

Do not multiply experimental conversion lift by the predicted LTV of observed buyers and label the product an established lifetime campaign effect. The tested population, outcome window and valuation assumptions must remain visible.

Parameter meanings and fitted estimates

Technical appendix | Original synthetic records; generation and future evaluation data are maintained separately.

LIKELIHOOD, HETEROGENEITY AND ECONOMIC CONVERSION

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$p_{i,g} \sim \mathrm{Beta}(\alpha_g, \beta_g), \qquad \alpha_g = \mu_g \phi_g, \qquad \beta_g = (1-\mu_g)\phi_g,$$
$$N_{g,t} \mid N_{g,t-1} \sim \mathrm{Binomial}(N_{g,t-1}, 1-h_g(t)), \quad h_g(t) = \frac{\alpha_g}{\alpha_g+\beta_g+t-1}.$$

g indexes a cohort–need–channel cell; $t = 1$ is its first renewal. Search and save-time customers are the reference groups. Transformations, estimation settings and uncertainty scope are documented on the next page.

Table 5. Structural parameter estimates, priors, posterior intervals and sampling diagnostics.

Parameter	Prior	Mean	5th–95th	R-hat	ESS
cohort_scale	HalfNormal(0.25)	0.121	0.062 to 0.187	1.000	1,114
concentration_beta[intercept]	Normal(log(4),1)	1.334	1.164 to 1.512	1.000	2,488
concentration_beta[build_skill]	Normal(0,0.5)	0.112	-0.118 to 0.342	1.001	2,895
concentration_beta[partner]	Normal(0,0.5)	0.075	-0.150 to 0.311	1.000	2,640
concentration_beta[organic]	Normal(0,0.5)	0.356	-0.353 to 1.084	1.000	5,879
mean_beta[intercept]	Normal(-2.5,1)	-1.684	-1.779 to -1.591	1.001	2,290
mean_beta[build_skill]	Normal(0,0.5)	-0.303	-0.414 to -0.195	1.002	3,068
mean_beta[partner]	Normal(0,0.5)	-0.214	-0.324 to -0.108	1.001	2,486
mean_beta[organic]	Normal(0,0.5)	-0.153	-0.548 to 0.226	1.000	5,204
mean_beta[cohort_trend]	Normal(0,0.5)	-0.350	-0.452 to -0.249	1.000	2,179
revenue_beta[intercept]	Normal(-0.5,0.8)	-0.605	-0.610 to -0.600	1.000	8,435
revenue_beta[log1p_age]	Normal(0.1,0.8)	0.108	0.105 to 0.110	1.000	6,064
revenue_beta[cohort_trend]	Normal(0,0.8)	0.021	0.017 to 0.026	1.000	7,731
revenue_beta[build_skill]	Normal(0,0.8)	0.057	0.053 to 0.061	1.000	7,646
revenue_beta[partner]	Normal(0,0.8)	0.001	-0.004 to 0.005	0.999	7,025
revenue_beta[organic]	Normal(0,0.8)	0.035	0.011 to 0.060	1.000	6,815
revenue_shape	LogNormal(log(4),0.8)	15.440	14.437 to 16.444	1.000	8,202

The 24 standardized cohort offsets z have posterior means from -1.529 to 1.088; multiply each draw by cohort_scale to obtain its log-odds shift. Their largest R-hat is 1.001; smallest effective sample size is 3,846. The complete parameter_summary.csv provides every cohort effect, prior, interval and interpretation.

At the coefficient means, Search / save time at centered cohort trend and zero offset has mean latent monthly churn mu=15.7% and concentration phi=3.80. The cohort-trend coefficient's 5th–95th interval is -0.452 to -0.249; it is below zero, supporting lower mean churn in later cohorts within this model. This is a model-based description, not a causal effect of acquisition date.

Posterior means and 5th–95th percentiles are read from the saved, hash-bound parameter output. Units and reference groups are stated with the parameter definitions; transformed parameters are not interchangeable with their latent log-scale values.

Document what the model estimates and what it assumes

Technical appendix | Parameter precision, predictive accuracy and causal identification are different properties.

ESTIMATION AND REPRODUCIBILITY

Table 6. Recorded inference settings and convergence diagnostics for the full-data fit.

Setting or diagnostic	Recorded value
Inference	NUTS
Chains / warmup / samples	4 / 1000 / 1000 per chain
Posterior forecast draws	400
Maximum R-hat / minimum ESS	1.002 / 1,114
Divergences	0
Seed / numerical backend	20261116 / cpu / float64

FEATURE TRANSFORMATIONS AND INTERPRETATION

Mean-churn coefficients use a logit link; concentration and revenue use log links. Search and save-time customers are the reference groups. Retention cohort trend is $(\text{index} - 11.5)/11.5$; revenue cohort trend is $(\text{index} - 11.5)/12$. Revenue log-age is $\log(1 + \text{age})$. Cohort index runs from 0 to 23. A coefficient therefore needs its link, scale and reference group before it has a managerial interpretation.

INTERVAL SCOPE AND REMAINING RISK

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval.

One historical year-end origin, not proof of accuracy for every future cohort or business regime.

> Convergence diagnostics assess whether the fitted posterior was explored adequately. They do not establish that the model captures future customer behavior or identifies a campaign's causal effect. Historical holdouts test prediction using information available at the forecast origin.

METHOD SOURCES AND ADAPTATION

Retention extends Fader-Hardie sBG with mean/concentration parameterization, logit/log covariate links, a non-centered cohort effect on mean churn, grouped conditional Binomial renewal likelihood and NUTS estimation. Revenue adapts Orduz's Gamma structure but uses observed active counts in fitting, an additional per-member shape multiplier, a mean link without an age-by-cohort interaction, need/channel covariates and $100 scaling. Conditional on observed activity, the revenue and retention likelihoods have independent parameter blocks. The separate Orduz-inspired 4-hidden-unit Bayesian neural network is a retention comparison only; its cumulative-count likelihood is changed to conditional renewal trials.

Show the unobserved renewal path explicitly

Technical appendix | The same model forecasts retained customers, then converts that path into revenue and contribution.

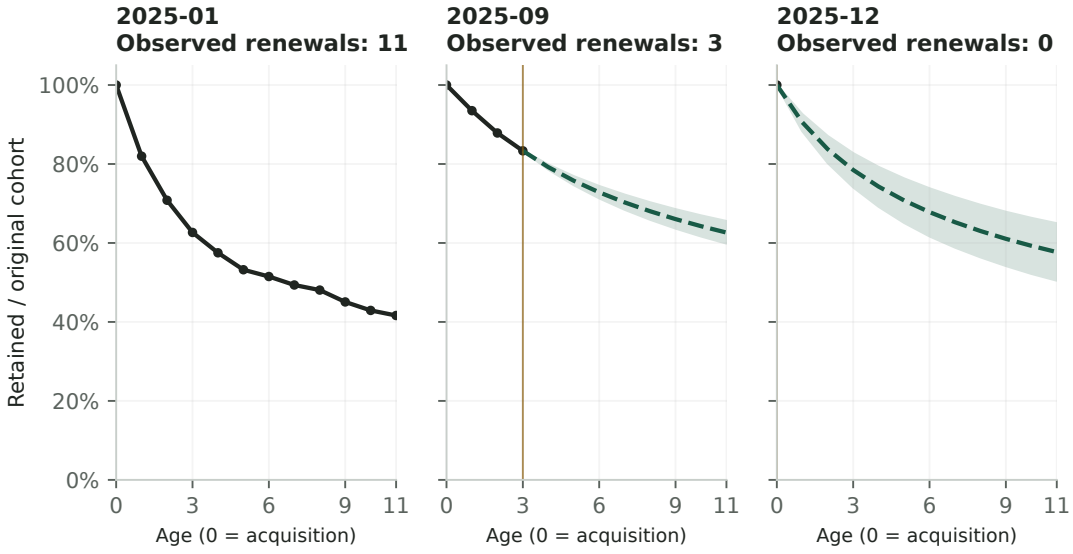


Figure 11. Observed and forecast retention for cohorts with complete, intermediate and minimal renewal histories.

Saved model output: `forecast_curve.csv`. Each panel uses its original acquired-customer denominator. Solid points are observed; dashed lines are forecast expected retention; shading is the 5th–95th posterior parameter range. The vertical marker is the observation boundary.

THE AMOUNT OF OBSERVED RENEWAL HISTORY MATTERS

The complete cohort supplies an observed reference. The intermediate cohort has some renewal history but still requires extrapolation. The newest cohort has only its acquisition month: it has **zero observed renewal decisions**. Its future path borrows information from older cohorts, stated needs, acquisition channels and the model's partially pooled cohort distribution.



Figure 12. Observed contribution and the forecast remainder for the newest acquisition cohort.

Observed contribution is retained; only the unobserved remainder is predicted. The newest cohort's forecast relies on older cohorts until its own renewal history becomes available.

> These are expected customer paths conditional on fitted parameters. The bands exclude future customer-level Bernoulli variation, uncertain economic conversion and changes in the business regime. A narrow parameter band is not evidence that those risks are small.

Separate fitted agreement from prediction performance

Technical appendix | A model can describe observed counts closely and still miss the future.

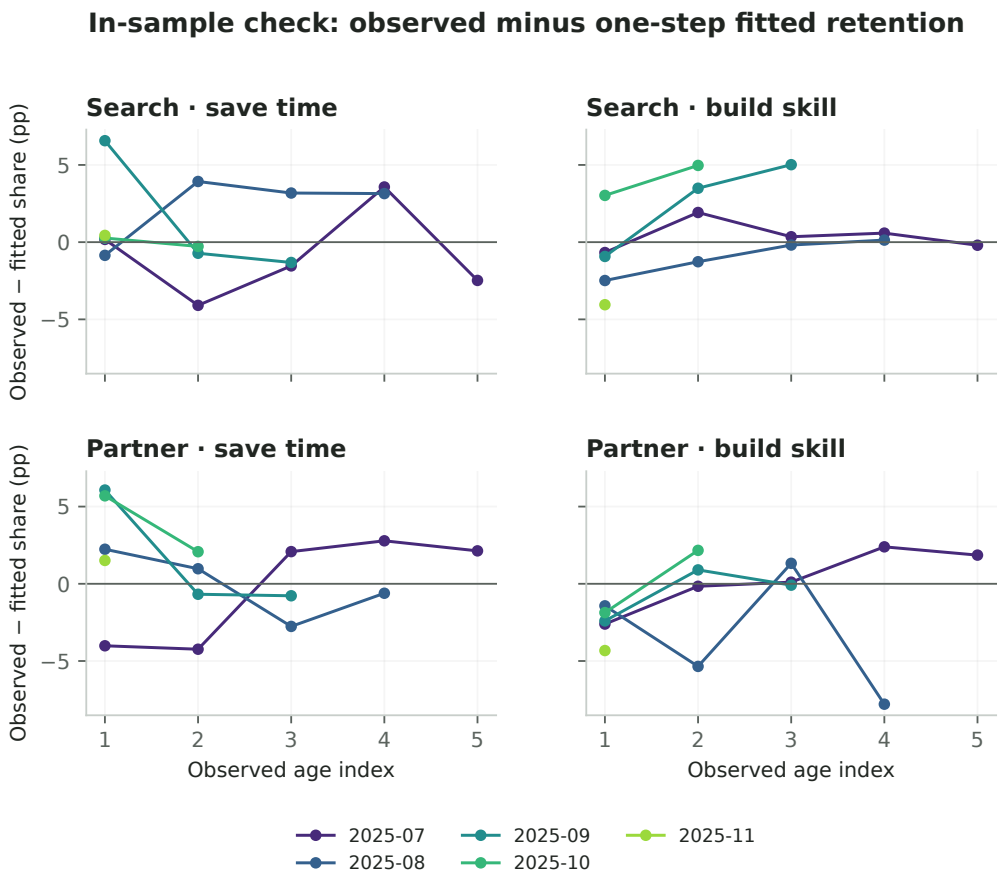


Figure 13. In-sample observed-minus-fitted retention residuals for the five newest cohorts with renewal history, by channel and need.

Saved SQL/model output: `model_fit_checks.csv`. These are in-sample, one-step fitted residuals conditional on previous observed customer counts. They are not held-out forecast errors.

WHERE CALENDAR-TIME FIT DEPARTS MOST

Table 7. The four largest absolute standardized calendar residuals among observed renewal risk sets.

Calendar month	Observed renewals	One-step fitted	Standardized residual
2025-03	1,569	1,609.6	-3.69
2025-04	1,654	1,678.6	-2.22
2024-05	530	544.9	-1.78
2025-02	1,510	1,528.1	-1.64

Four months with the largest absolute standardized calendar residual, selected from saved `calendar_fit_checks.csv`; only renewal ages 1 and older enter, not new acquisitions. These flag timing patterns, not causes, held-out errors or formal significance tests.

Test the forecast on outcomes withheld from fitting

Technical appendix | Historical validation is separate from the private future competition holdout.

HELD-OUT PERFORMANCE

Training includes 2025-09; withheld months are 2025-10, 2025-11, 2025-12.

Withheld cells: 258. Retained-count mean absolute error: 1.29. Revenue weighted absolute percentage error: 5.7%. A separate training-only forecast checks complete 12-month values: LTV mean absolute error $2.87 versus $23.43 for its baseline. These tests evaluate prediction, not campaign effects.

Table 8. Prediction performance on historical holdouts, with explicit baseline comparisons and units.

Measure	Model	Baseline	Interpretation
Retained-count MAE	1.29	2.63	People per withheld cell
Retained-count WAPE	4.5%	Not reported	Retained customers, not revenue
Revenue WAPE	5.7%	12.3%	Withheld calendar months
12-month LTV MAE	$2.87	$23.43	Value per original customer

On the same 258 held-out cells, retained-count MAE is 1.291 for sBG, 1.180 for the Orduz-inspired 4-hidden-unit network, and 2.633 for the naive baseline. The network uses variational inference and a conditional monthly churn likelihood; it does not supply executive LTV. Revenue WAPE evaluates the main Gamma revenue model, not the retention-only network.

WHAT THE COMPARISON CAN ESTABLISH

The model and each baseline use the same withheld outcome cells. Compare their errors in the stated units and at the same forecast horizon. A better retained-count forecast need not produce the better revenue forecast, and predictive performance does not establish a campaign's causal effect.

> The future challenge covers the next 12 calendar months for the already acquired cohorts. This differs from the executive report's valuation horizon of each cohort's first 12 monthly receipts. Hidden future outcomes are reserved for scoring and do not enter these historical checks.

Show exactly how much of each value was forecast

Technical appendix | Training includes the calendar month beginning 2024-12-01; later calendar months are used only for scoring.

At the 2024-12 forecast origin, 11 cohorts had an unfinished 12-month value horizon. Payback had already occurred for 7; those cases are excluded from forecast-month MAE. The 4 remaining cases with both observed and predicted recovery have payback-month MAE 0.00. This is a small future-payback sample, not evidence of universally exact prediction.

Table 9. Cohort-level historical value and payback forecasts, separating already observed recovery from forecast recovery.

Cohort	Observed / forecast months	Actual LTV	Model LTV	Baseline LTV	Payback actual / model	Payback evaluation
2024-02	11 / 1	$177.81	$177.30	$177.49	7 / 7	Already observed
2024-03	10 / 2	$202.82	$202.37	$202.53	6 / 6	Already observed
2024-04	9 / 3	$169.35	$171.34	$170.61	7 / 7	Already observed
2024-05	8 / 4	$191.75	$189.54	$187.20	6 / 6	Already observed
2024-06	7 / 5	$190.76	$188.79	$181.60	6 / 6	Already observed
2024-07	6 / 6	$202.68	$203.77	$195.59	5 / 5	Already observed
2024-08	5 / 7	$202.42	$198.31	$178.29	5 / 5	Already observed
2024-09	4 / 8	$207.28	$208.34	$177.16	5 / 5	Future; scored
2024-10	3 / 9	$198.53	$202.04	$155.32	5 / 5	Future; scored
2024-11	2 / 10	$208.93	$211.65	$160.56	5 / 5	Future; scored
2024-12	1 / 11	$202.00	$213.95	$112.78	5 / 5	Future; scored

Saved model output: `valuation_backtest.csv`. LTV values are discounted contribution before blended CAC, per original acquired customer. “Observed / forecast” counts the receipts within the common 12-month horizon; payback month counts receipts from 1.

WHY TOTAL-VALUE AND FORECAST-TAIL ERRORS ARE THE SAME

Observed contribution is copied identically into the actual and predicted totals. The forecast covers only the remaining months. Consequently, total LTV error equals the error in the unobserved tail; the long observed prefix does not create additional predictive accuracy. Rows are ordered by the amount of history available at the origin.

THE BASELINE AND ITS WEAKEST-INFORMATION CASE

The naive baseline freezes the last three observed renewal rates and revenue per retained customer. A cohort with only its acquisition month has no observed renewal rate; the baseline uses an explicit 0.8 renewal fallback. Its value is an assumption, not an estimated renewal rate. The December row makes this low-information comparison visible.

WHAT THE PAYBACK FORECAST REPRESENTS

Forecast payback is the first crossing of acquisition cost by the **posterior-mean expected contribution path**. It is not the mean or median of a payback-time distribution, and it does not estimate the chance that a realized customer cohort recovers its cost by that month. Posterior parameter bands also exclude future customer-level process noise.

> The inference model is an approximation to observed customer behavior, not access to the process that created the source records. The small set of future-payback cases is a narrow diagnostic: interpret zero month error, if observed, alongside full value errors and incomplete histories. The future competition holdout remains private and is not part of these historical validation results.

Preserve source meaning through the joins

Technical appendix | Eight separate exports become tested analytical views in DuckDB; no database server is required.

Eight source exports; 24 monthly acquisition cohorts; cutoff 2026-01-01. Pending renewal outcomes are excluded from observed renewal trials. All randomized prospects remain in experimental denominators, including nonbuyers.

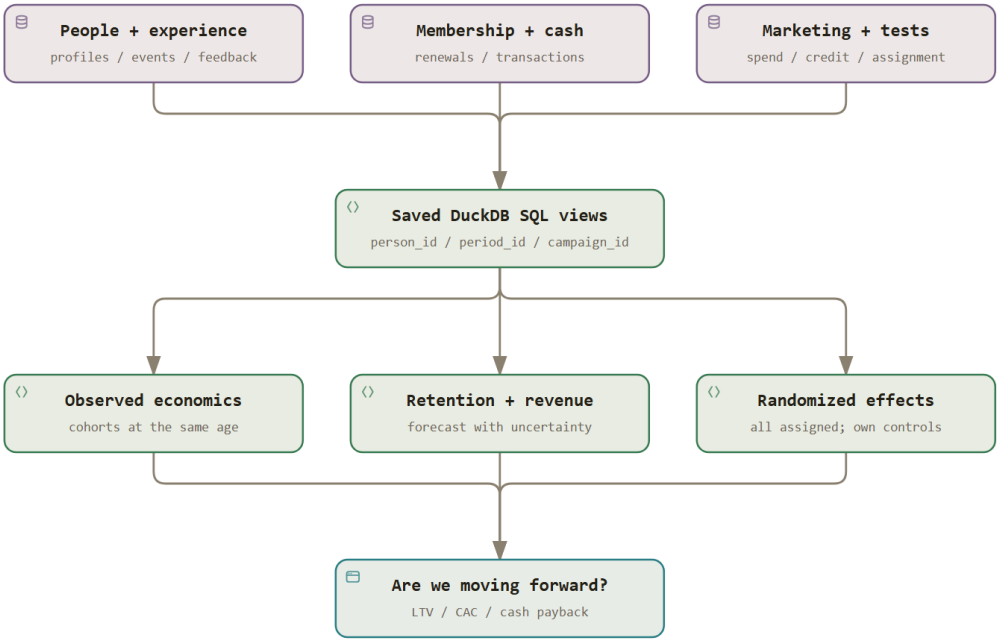


Figure 14. Eight source exports become analytical views and decision evidence through an auditable local pipeline.

The diagram is rendered by the bundled Archify package from original source specifications. The accompanying `diagrams/source-tables.html` and SVG show the full table-level schema and cardinality. Dataset classifications on the following pages list concrete fields.

KEYS DO NOT MAKE UNLIKE RECORDS INTERCHANGEABLE

A person can have many renewal periods, ledger transactions, product events and survey responses. Those tables must be aggregated at the intended grain before they are joined. Acquisition spending is a campaign-level record; it does not become a customer-level observation merely because it can be allocated across acquired customers. Experimental assignment defines an eligible population that includes noncustomers.

ANALYTICAL CONTRACTS

Table 10. Source-to-analysis definitions that preserve grain, timing, denominator and cost scope.

Definition	Contract
Joining cohort	Month of the first successful payment, derived from complete payment history.
Customer survival	Renewing contractual membership; original cohort size is the retained-share denominator.
Revenue and contribution	Ledger payments less discounts/refunds; contribution also subtracts recorded variable costs.
Acquisition investment	Tracked spend, including campaigns with no attributed acquisitions; no invented organic-source CAC.
Experimental unit	Every assigned eligible prospect, including nonbuyers and control buyers, within the prespecified follow-up window.
Unknown future	Right-censored ages stay unobserved. Missing survey responses remain missing perceptions.

Source hashes, saved SQL and downstream manifests bind the evidence chain. Numeric tables, editable LaTeX, source diagrams and the maintained style package accompany the report.

Source records, keys and cardinality

A3 landscape foldout | The eight raw exports remain separate. Campaign metadata is a derived lookup for a nonunique campaign-spend key.

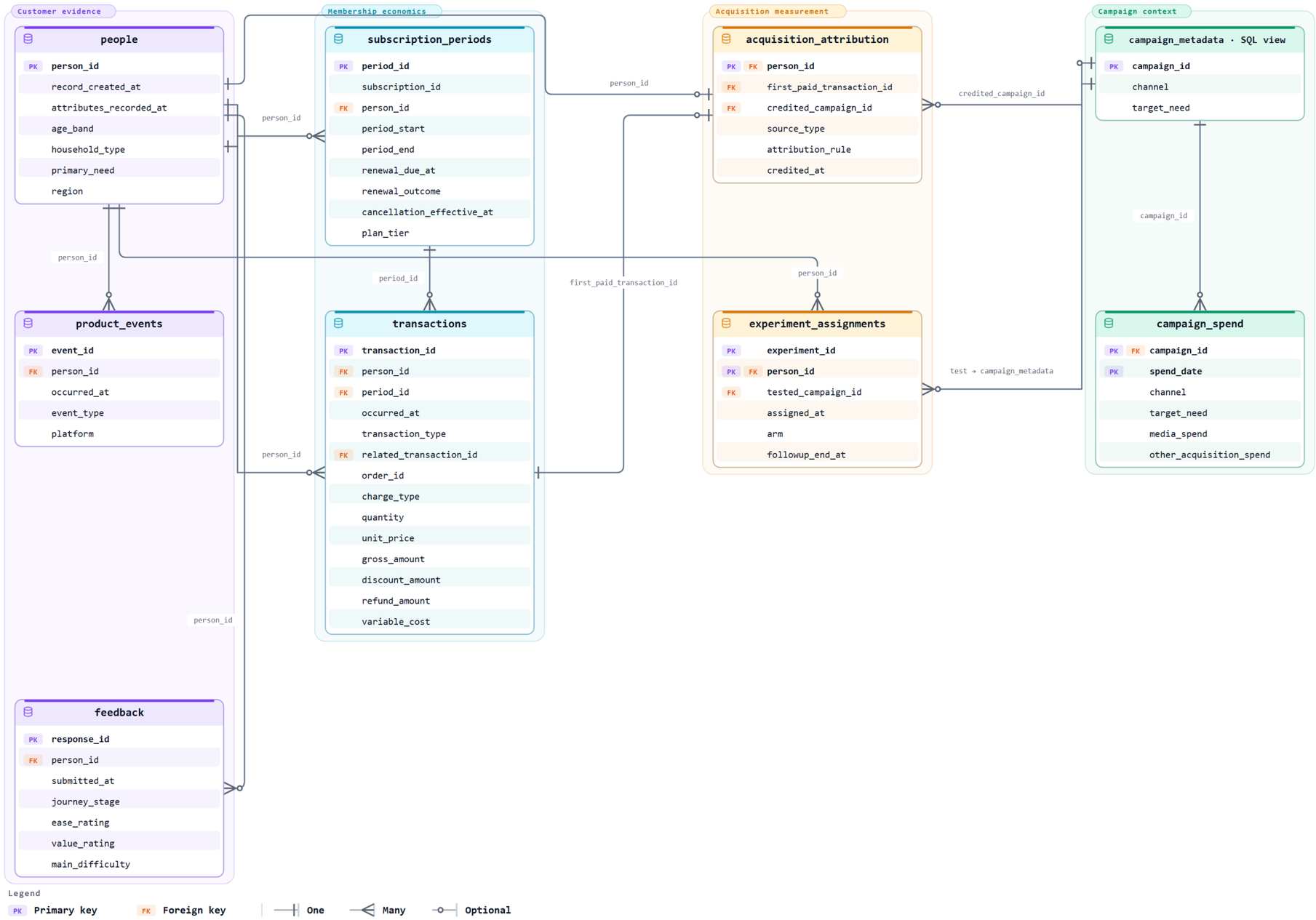


Figure 15. Source tables, join keys and cardinality, including the derived unique campaign lookup.

Native Archify schema; editable SVG/HTML: diagrams/source-tables. Refunds link transactions.related_transaction_id to transactions.transaction_id within the same ledger. Campaign foreign keys target unique campaign_metadata; campaign_spend uses campaign_id + spend_date. Aggregate to the intended grain before joining one-to-many records.

Classify each source by what it actually measures

Technical appendix | Context is marked explicitly; the categories describe evidence, not proof of an individual action or attitude.

Identifiers: Keys and profile descriptors identify or link people, groups and records; not every identifier is unique. Engagement: Observed customer interactions and their measurement context. Firm marketing activity and attribution metadata are retained as explicitly marked contextual evidence; they do not establish individual exposure, action or a causal effect. Relationship: Contractual and commercial arrangements, customer status, and explicitly marked operational context for relationship economics. Perception: Customer-reported views or intentions, distinguished from behavior and analyst inference.

Table 11. Universal Customer View: people.csv.

CRM | One person, including eligible prospects who never purchase.

Identifiers	Engagement	Relationship	Perception

`person_id`
`age_band`
`household_type`
`region`
	Not observed	`record_created_at`†
		`attributes_recorded_at`†
			`primary_need`

record_created_at: CRM relationship provenance; registration does not establish a paid customer. attributes_recorded_at: Profile provenance: verifies attributes were available before acquisition or assignment.

Table 12. Universal Customer View: subscription_periods.csv.

Renewal system | One scheduled monthly cycle; a paid outcome establishes retained membership for that cycle.

Identifiers	Engagement	Relationship	Perception

`period_id`
`subscription_id`
`person_id`
	Not observed	`period_start`
		`period_end`
		`renewal_due_at`
		`renewal_outcome`
		`cancellation_effective_at`
		`plan_tier`
			Not observed

Table 13. Universal Customer View: transactions.csv.

Finance | One payment or refund event; each payment is one order and refunds reuse its order ID.

Identifiers	Engagement	Relationship	Perception

`transaction_id`
`person_id`
`period_id`
`related_transaction_id`
`order_id`
	`occurred_at`
	`transaction_type`
	`charge_type`
	`quantity`
	`gross_amount`
	`discount_amount`
	`refund_amount`
		`unit_price`†
		`variable_cost`†
			Not observed

unit_price: The item's commercial price term is Relationship context; billed quantity and realized amounts record transaction Engagement. variable_cost: Firm-side accounting context for relationship economics; unlike billed amounts, it does not record a customer payment or reported perception.

Each raw field appears once in its dataset's classification. Source-specific rationale is preserved in `diagrams/ucv-tables.json`. A category marked not observed is a data gap, not an invitation to infer a missing attitude or relationship.

Keep campaign credit, response and belief distinct

Technical appendix | Campaign spending and randomized assignment describe different objects from a customer's behavior or survey response.

Table 14. Universal Customer View: product_events.csv.

Product and support | One observed product/support interaction; no event does not mean no subscription.

Identifiers	Engagement	Relationship	Perception

`event_id`
`person_id`
	`occurred_at`
	`event_type`
	`platform`
		Not observed	Not observed

Table 15. Universal Customer View: feedback.csv.

Customer research | One completed linked survey response; nonresponse is missing, not a zero rating.

Identifiers	Engagement	Relationship	Perception

`response_id`
`person_id`
	`submitted_at`
	`journey_stage`†
		Not observed	`ease_rating`
			`value_rating`
			`main_difficulty`

journey_stage: Context of the response, not a measured universal customer path.

Table 16. Universal Customer View: campaign_spend.csv.

Marketing spending | One campaign per spending date; aggregate before joining to customer outcomes.

Identifiers	Engagement	Relationship	Perception
`campaign_id` `target_need`†	`spend_date`† `channel`† `media_spend`† `other_acquisition_spend`†	Not observed	Not observed

target_need: Identifies the campaign's target group; unlike a person's primary_need, it is not an individually stated intention. Channel describes the firm's marketing activity, so it is Engagement context instead. spend_date: Timing of firm marketing activity, not a customer touchpoint timestamp. channel: Context of firm marketing activity; it does not prove an individual's exposure. media_spend: Firm-side campaign spending is retained as Engagement context, not individual engagement or evidence of exposure. other_acquisition_spend: Firm-side creative/vendor costs are Engagement context, not observed customer activity; aggregate once before joining customers.

Separate marketing credit from experimental assignment

Technical appendix | A dagger marks contextual placement. It is not part of a source field's name.

Table 17. Universal Customer View: acquisition_attribution.csv.

Marketing attribution | One acquired person and the recorded credit for the first successful payment.

Identifiers	Engagement	Relationship	Perception
person_id first_paid_transaction_id credited_campaign_id	source_type† credited_at† attribution_rule†	Not observed	Not observed

source_type: Recorded acquisition-path credit; an attribution label does not establish incrementality. credited_at: When credit was recorded, not necessarily the acquisition or exposure time. attribution_rule: Measurement metadata for engagement credit, not a customer action itself.

Table 18. Universal Customer View: experiment_assignments.csv.

Experiment platform | One eligible person per randomized experiment, including all nonbuyers.

Identifiers	Engagement	Relationship	Perception
experiment_id person_id tested_campaign_id	Not observed	assigned_at† arm† followup_end_at†	Not observed

assigned_at: Experimental relationship/design metadata; assignment is not observed exposure or purchase. arm: Randomized intervention assignment; control nonbuyers must remain in the denominator. followup_end_at: Prespecified measurement window, not a subscription end date.

SOURCES AND MODEL ADAPTATIONS

Juan Orduz, Revenue Retention and Customer Lifetime Value with NumPyro; Peter Fader and Bruce Hardie, contractual-customer retention and shifted-Beta-Geometric models; Gil Bouhnick, Cohort Analysis. Exact links, adaptation boundaries and licensing are retained in the repository source documentation. Archify is pinned and redistributed with its license notices.

Retention extends Fader-Hardie sBG with mean/concentration parameterization, logit/log covariate links, a non-centered cohort effect on mean churn, grouped conditional Binomial renewal likelihood and NUTS estimation. Revenue adapts Orduz's Gamma structure but uses observed active counts in fitting, an additional per-member shape multiplier, a mean link without an age-by-cohort interaction, need/channel covariates and $100 scaling. Conditional on observed activity, the revenue and retention likelihoods have independent parameter blocks. The separate Orduz-inspired 4-hidden-unit Bayesian neural network is a retention comparison only; its cumulative-count likelihood is changed to conditional renewal trials.

All graphics in this report are generated from this repository's original synthetic records or original diagram specifications. The public sources motivate analytical ideas; their screenshots and private course materials are not redistributed.

Make the analytical handoff executable and reproducible

Skills specify decisions and stage gates; reference assets explain methods; Python packages execute the transformations and models.

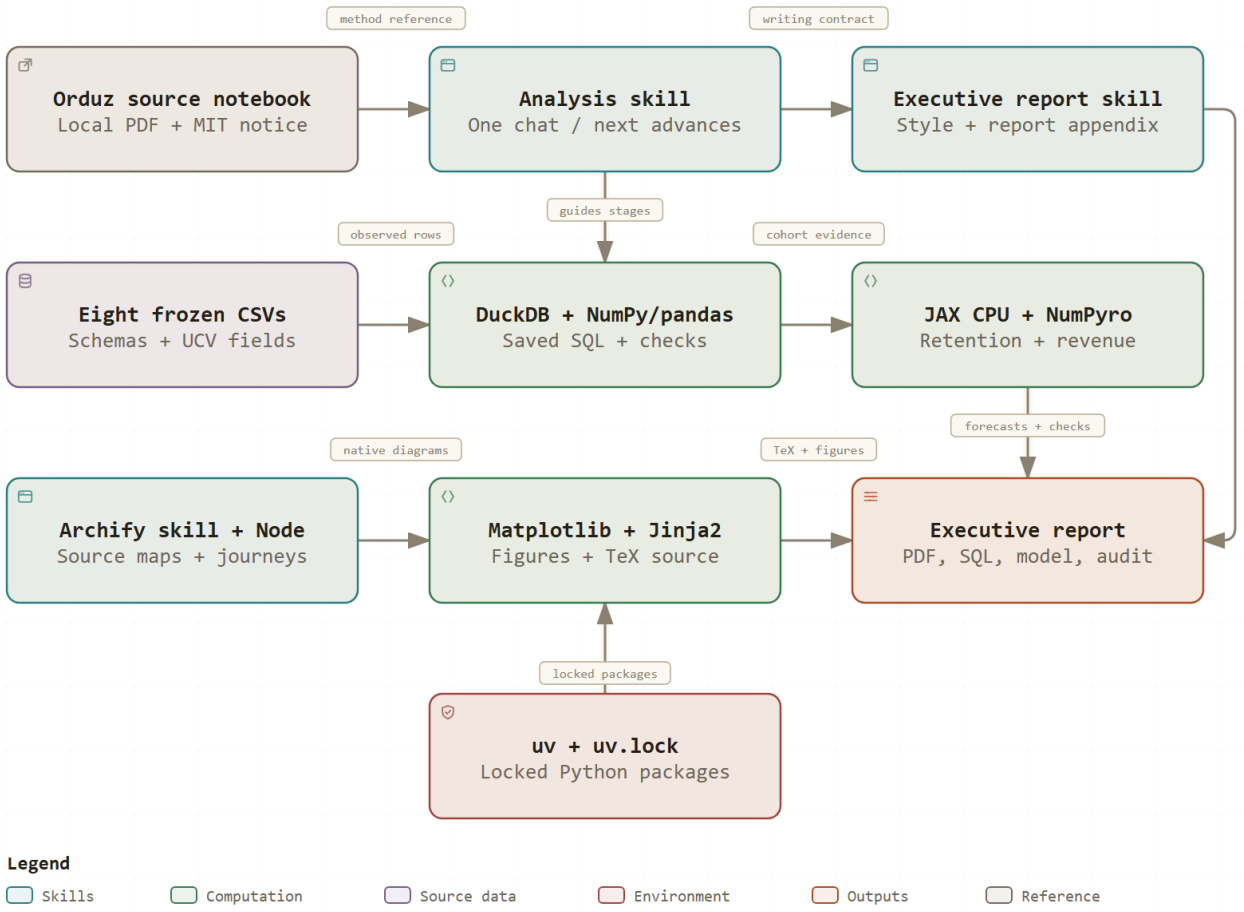


Figure 16. Knowledge assets and execution dependencies. Typed boxes, icons and colors distinguish skills, reference documents, Python packages, data and generated outputs.

THE SAME EVIDENCE SHOULD LEAD TO THE SAME REPORT

Use `uv sync --frozen` to install the recorded Python environment and `uv run --frozen` to execute its commands. The model uses CPU JAX/NumPyro; a GPU is not required. The analysis begins with the released records and does not regenerate them. Source hashes, saved SQL, model settings, figure data and document assets remain bound to the output manifests.

> The student workflow does not contain the private data-generation skill or the future holdout. A challenge model must make its predictions from the same observed information. Hidden future outcomes are reserved for evaluation, not feature construction, parameter tuning or revising the input data.